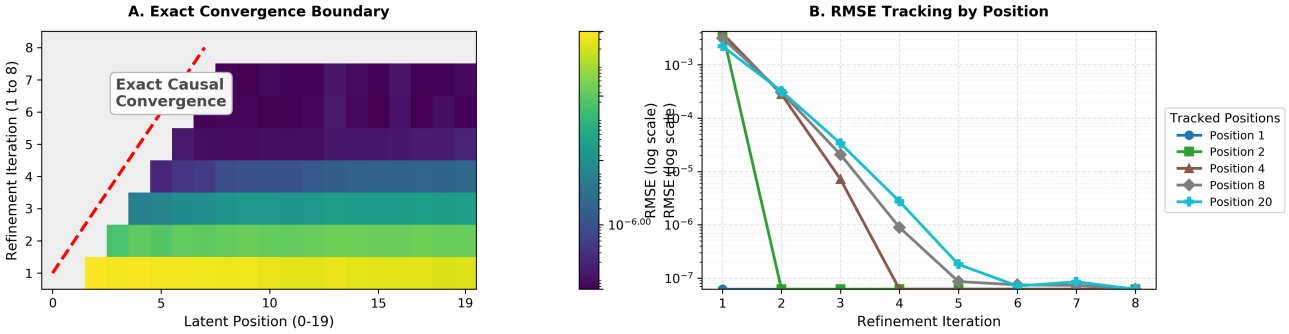

Figure 6. **Position-level convergence of Jacobi iteration on LLaMA-3-3B CPT. (A)** Heatmap of per-position RMSE between the Jacobi iterate $H^{(k)}$ and the sequential fixed point H_{seq} across refinement iterations (y-axis) and latent positions (x-axis). The red dashed line marks the exact causal convergence boundary: positions to its left have converged to machine precision. Earlier positions converge first, forming a clear left-to-right front consistent with the autoregressive causality argument (Appendix D.1). **(B)** RMSE trajectories for selected positions. Position 1 converges after 1 iteration, Position 2 after 2, while Position 20 requires $\sim$7 iterations. All positions reach machine precision ($< 10^{-7}$), confirming rapid and exact convergence.